# Does the weekly focus's grade tell a real improvement from none?

A replay of the ONE focus on the player's games, and a simulation with a known truth

Chess Coach research lab

**Abstract**

Every Sunday the weekly review picks one focus, the leak cause where the player gives away the most against his opponents, freezes a baseline and a target, and grades it over the next 40 games. A met focus is the plan's evidence that the training worked. Replayed on his archive, the picker changed focus eight times in ten weeks and graded five of seven finished focuses met. In 4,000 simulated focuses drawn from his own games with nothing changed, the grade says met 44% of the time at 40 games and 55% at 120, against the 10% its own 0.9 threshold implies; a cause drawn at random gets 36%, so most of it is the test, not the choice. A permutation test against older games that played no part in the choice holds its level (8.3% and 9.1%), and reading points per move of the cause sees a real cut sooner than points per game. The targets ask for a median cut of 8% while 40 games can only see about 31%. A candidate grade built on those findings says met 9.5% of the time with nothing changed and 61% with a quarter of the loss removed, over 120 games; on his games it graded not met a focus the app had graded met on a baseline read at its worst.

## 1 The question

The weekly's ONE focus is the app's answer to "what should he work on now". It ranks the leak causes (an opening, middlegame or endgame position that is tense or quiet, under a minute on the clock, a fast move in a tense position, the move after a gift, late in a game) by the Elo they cost, the probability the leak is real, its relevance, its actionability and its freshness, and keeps a running focus unless a challenger scores 1.5 times as much. A chosen focus is pre-registered: a baseline (his points lost on the cause per game over the last 100 games), a target halfway to his opponents' level, and a CUSUM test. It is graded over the next 40 games: met when the CUSUM alarms downward or the posterior probability that his loss fell reaches 0.9, else not met.

Two things decide whether that grade can be read as evidence: how often it says met when nothing changed, and how large a change it needs before it says met. Both are measured here on the app's own code at commit `d8b771238c`, through the lab's bridge.

## 2 What the app did with his games

The replay runs the weekly's own choice and commit every Sunday of the archive (634 games), on a metrics instance that holds only the games played by then. He plays a median of 50 games a week, more than the 40-game window, so in eight of the nine weeks that had a focus running, it had already been decided by the next Sunday, and the picker moved on: eight focuses in ten weeks. Of the seven that finished, five were graded met and two not met. At the simulated null rate below, five or more of seven would read met by chance alone 14% of the time.

The picker's probability that a leak is real averaged 40% over the 90 ranked leaks of the replayed weeks, and 40% of them were still there (his loss on the cause still above his opponents') over the next 40 games. It did not say which: its Brier score for that outcome was 0.246, against 0.240 for a constant. A 40-game future is noisy, so this does not show the probability is wrong; it shows that, over the span a focus runs, it does not tell the leaks that will show up again from those that will not. One switch went to a leak the picker gave even odds (0.50), with a gap of 0.03 points a game against a standard error of 3.19.

## 3 A simulation with a known truth

Each of 4,000 runs draws, with replacement from his 634 archived games, an older 100-game history, the 100 games the picker reads, and a window of up to 120 games after it, so nothing changes between them. The app's `rankFocus`, `preregister` and `gradeFocus` run on the draws exactly as the weekly would. The same draws are then graded with a known share of his loss on the chosen cause removed: each of his moves on the cause loses nothing with that probability, from a fixed uniform per move, so every cut is compared on the same games. Beside the app's grade, four candidates were scored on the same draws (fixed before this run, after a first run measured the app's grade alone): the app's grade on a cause drawn at random from the ranked ones (no selection), the app's grade without its CUSUM pass, and a one-sided permutation test (400 permutations, level 10%) of the window against the *older* games, which played no part in choosing the cause, on points per game and on points per move of the cause.

**With nothing changed.** The app's grade says met 44% of the time at 40 games, and more as the window grows (55% at 120): the CUSUM alone fires 36% of the time and the posterior alone 28%. On a cause drawn at random the grade still says met 36% of the time (the posterior alone 19%), so selection adds little; the test itself is the problem. His per-game loss on a cause is heavily skewed (a median coefficient of variation of 1.5): most games lose little and a few lose a lot, so a short window usually sits below the mean, and a CUSUM tuned to a normal-looking shift alarms on it. The permutation tests, which make no such assumption, hold their level.

**The choice inflates the baseline.** With nothing changed, the window's loss on the chosen cause sits a median 10% below the baseline that chose it, and the older games sit 9% below it too: the picker reads the baseline at its worst. A test against the baseline credits that regression to the mean to the training. The older games carry no such bias.

**When something changed.** With a quarter of his loss on the cause removed, the app's grade says met 71% of the time at 40 games, but 44% of that it would say anyway: net 27 points. Points per move against the older games say met 44% of the time (net 35 points), 61% at 120 games; with half of it removed, 76% and 85%. Per move beats per game at every cut and window, because a game with many moves on the cause and a game with one are no longer weighed alike.

**Targets the window can see.** The pre-registered target asks for a median cut of 8% of his loss on the cause (from 2% at the tenth percentile to 38% at the ninetieth). For the posterior to reach 0.9 on 40 games, the cut has to be about 31% at the median spread. Most targets are below what the window can tell from noise.

| grade, 40 games | no change | 10% cut | 25% cut | 50% cut |
|---|---|---|---|---|
| the app | 44.3 | 55.3 | 71.1 | 87.6 |
| its CUSUM alarm alone | 35.9 | 45.4 | 59.3 | 74.4 |
| its posterior alone | 28.1 | 39.5 | 60.8 | 84.8 |
| fresh (per game) | 8.3 | 15.5 | 31.1 | 63.2 |
| fresh, per move | 9.1 | 20.0 | 44.1 | 76.3 |
| random cause, the app's grade | 35.5 | | | |

| grade, 80 games | no change | 10% cut | 25% cut | 50% cut |
|---|---|---|---|---|
| the app | 51.9 | 65.0 | 80.6 | 94.0 |
| its CUSUM alarm alone | 44.8 | 55.9 | 69.7 | 83.7 |
| its posterior alone | 31.4 | 48.4 | 72.8 | 92.5 |
| fresh (per game) | 10.0 | 19.0 | 41.3 | 74.0 |
| fresh, per move | 9.8 | 23.5 | 54.9 | 82.4 |
| random cause, the app's grade | 41.8 | | | |

| grade, 120 games | no change | 10% cut | 25% cut | 50% cut |
|---|---|---|---|---|
| the app | 55.0 | 69.1 | 84.0 | 95.2 |
| its CUSUM alarm alone | 46.0 | 57.5 | 72.0 | 86.3 |
| its posterior alone | 35.4 | 54.6 | 77.9 | 94.8 |
| fresh (per game) | 10.0 | 20.4 | 46.7 | 78.2 |
| fresh, per move | 9.6 | 25.6 | 60.7 | 84.8 |
| random cause, the app's grade | 43.6 | | | |

Table 1: The share of simulated focuses graded met, by grade, window and the share of his loss on the cause removed. With no change, every share above the level (10%) is a false met.

**Not met is mostly "cannot tell".** Of the focuses graded not met with nothing changed, 87% had a posterior between 0.1 and 0.9: the grade had no evidence either way, but the weekly records it as not met.

## 4 What it means for the app

The grade is the evidence the plan reads, so it should earn its word. The simulation points to five changes, each measurable again on this harness before it ships:

1. Grade a focus against games that did not choose it (the games before the baseline, or a held-out half), not against the baseline that picked it.
2. Drop the CUSUM early pass, or calibrate its threshold by resampling his own games, so a skewed loss does not alarm by itself.
3. Read points per move of the cause, not per game.
4. Say three things, not two: improved, not improved, and not enough games yet.
5. Size the window, or hold the focus, to the cut the target asks for: at his pace a focus can run two or three weeks of games instead of one.

## 5 A candidate grade

The candidate (`candidates/permove/focus.ts`) is the app's `focus.ts` with the first four changes and a longer hold. It grades a focus by a one-sided permutation test, 1,000 shuffles of games seeded by the focus, of his points per move on the cause in the window against the 100 games before the ones the choice read (the baseline itself when the archive is shorter). It says met when the test's p-value is at most 0.1, not met when it is 0.5 or more (the window is no better than the older games), and unclear in between, and it holds a focus for 120 games with no early pass. It ran through the same simulation on the same draws, again on fresh draws (seed 2), and through the same weekly replay.

**It holds its level.** With nothing changed, it says met 9.7% of the time at 40 games, 9.4% at 80 and 9.5% at 120, and 10.1% on fresh draws; on a cause drawn at random, 9.2%. On the fresh draws the app's grade says met 44% of the time at 40 games, so its false met is not a property of one set of draws.

**It sees a real cut.** With a tenth, a quarter and a half of his loss on the cause removed, it says met 25%, 61% and 85% of the time at 120 games (45% at 40 for a quarter). These are the per-move test's rates above, within two points at every cut and window: the candidate is that test, with its own shuffles.

**What the weekly would say.** At the end of a 120-game window with nothing changed, the candidate says met 9.5%, unclear 39.4% and not met 51.1% of the time; with a quarter of the loss removed, 61.1%, 29.3% and 9.6%. Not met now means something: it is common when nothing changed and rare when something did.

**On his games.** Replayed over the same ten weeks, the candidate held four focuses instead of eight, each for two to four weeks. It graded two of them unclear and one not met, and none met. The one it graded not met, `middlegame.quiet`, is one the app graded met: his loss on the cause fell from 17.3 points a game in the baseline to 10.9 in the 40 games after it, but the 100 games before the baseline had read 10.3. Per move, the windows after the choice (3.70 over 40 games, 3.88 over 120) were no better than the older games' 3.61, while the baseline that chose it read 4.69. The app's met was the regression to the mean the simulation predicts.

## 6 Limits

The simulation draws games independently, so it has no streaks, no rating drift and no change in his opponents; a real improvement may arrive unevenly. The cut removes loss uniformly across his moves on the cause, which is one shape of improvement among many. The replay covers ten weeks, too few to grade the picker's choices on outcomes; it shows how the loop behaves, not whether it helped. The candidate compares a window with games played weeks earlier, so anything else that changed in between (his openings, his opponents, his rating) reads as a change too.